

AutoGen 0.4: A New Era of AI Agents



Ground-Up Rewrite

A fundamental architectural shift for the AI agent ecosystem.



Beyond Simple Chat

Designed for complex, large-scale multi-agent topologies.



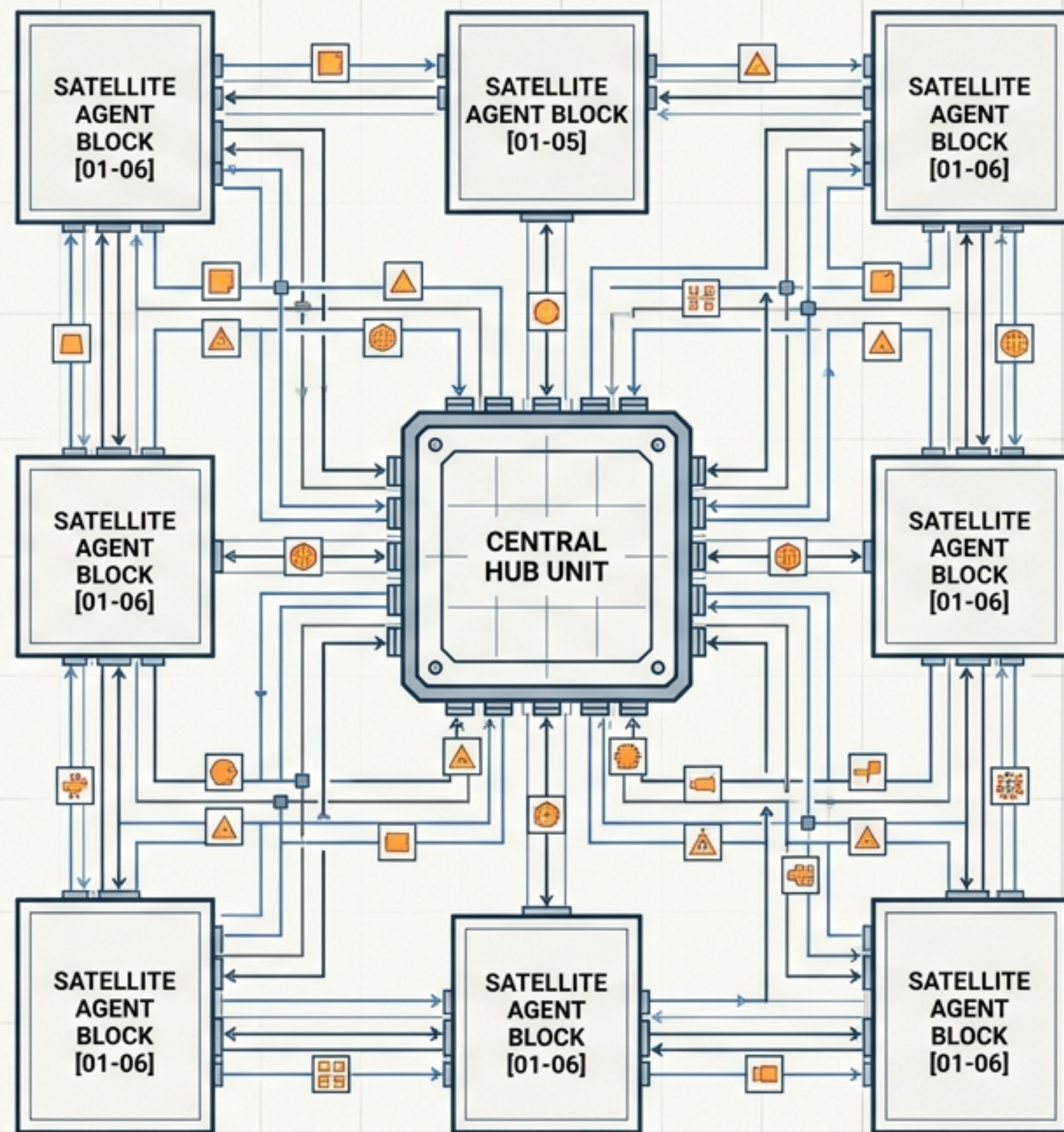
Event-Driven Core

Modular, asynchronous, and fully decoupled.



Enterprise Ready

Engineered specifically for robust production deployments.



Breaking Past Limitations

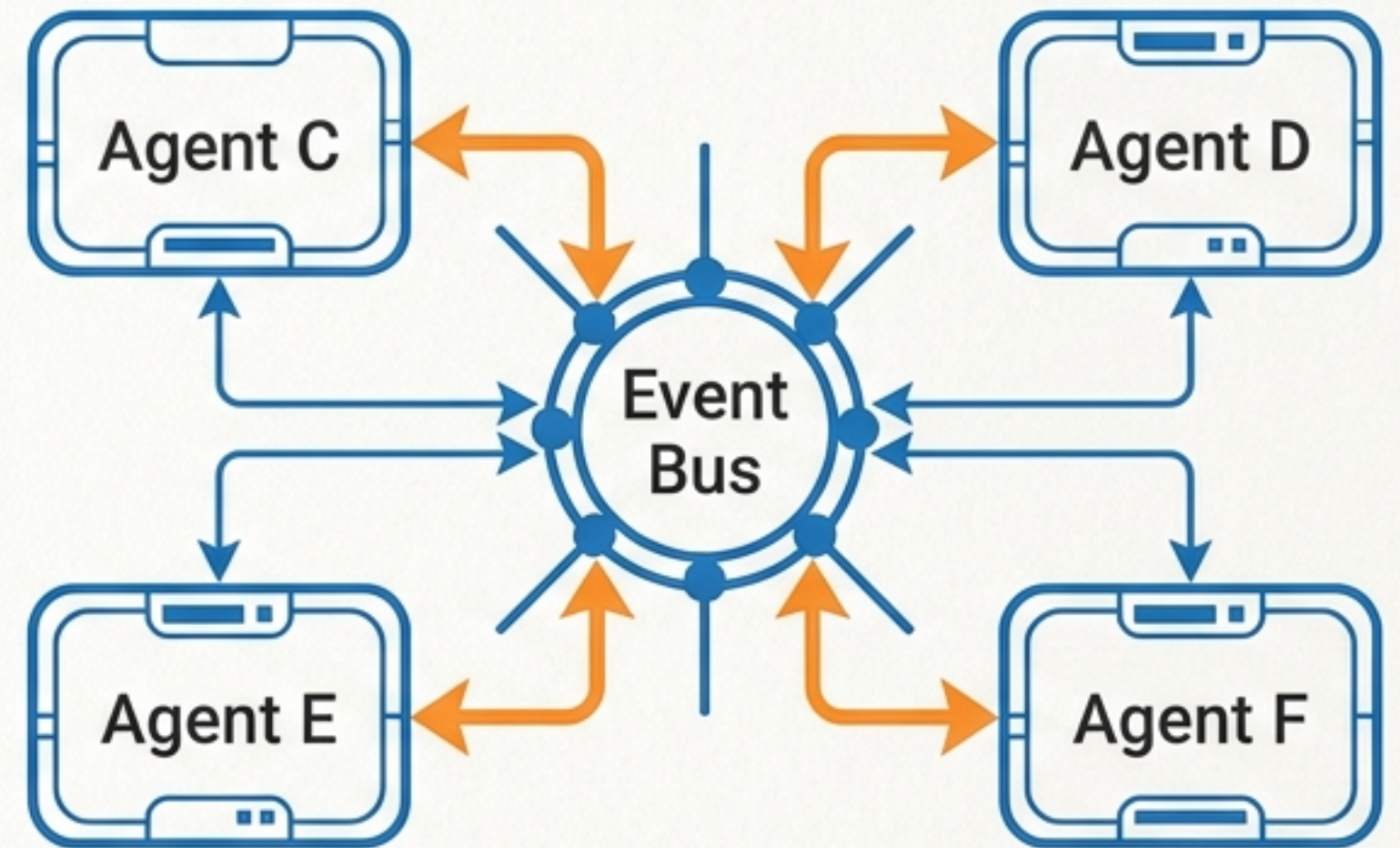
Old Frameworks



Tightly Coupled

Sequential and
Blocking

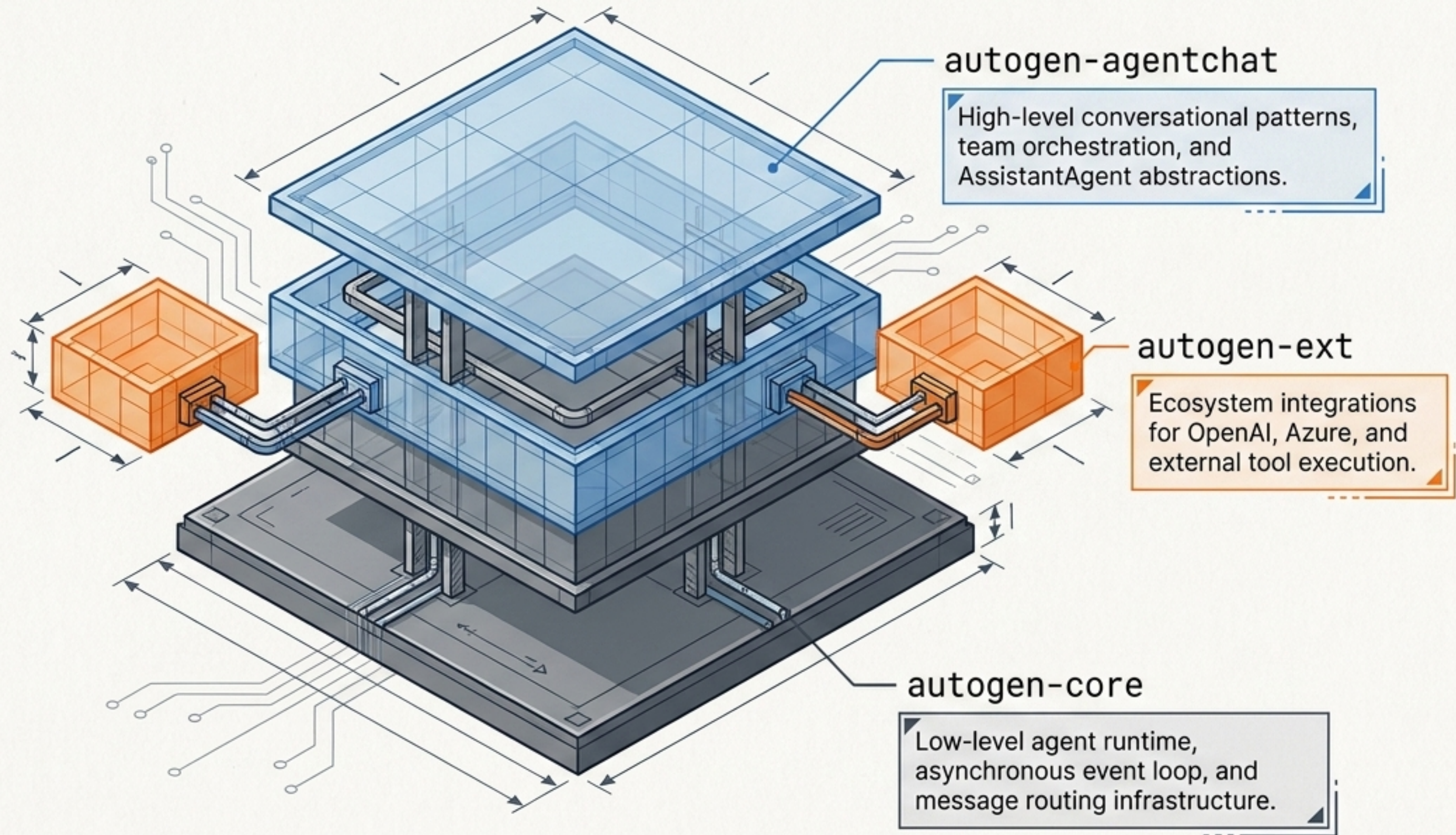
AutoGen 0.4



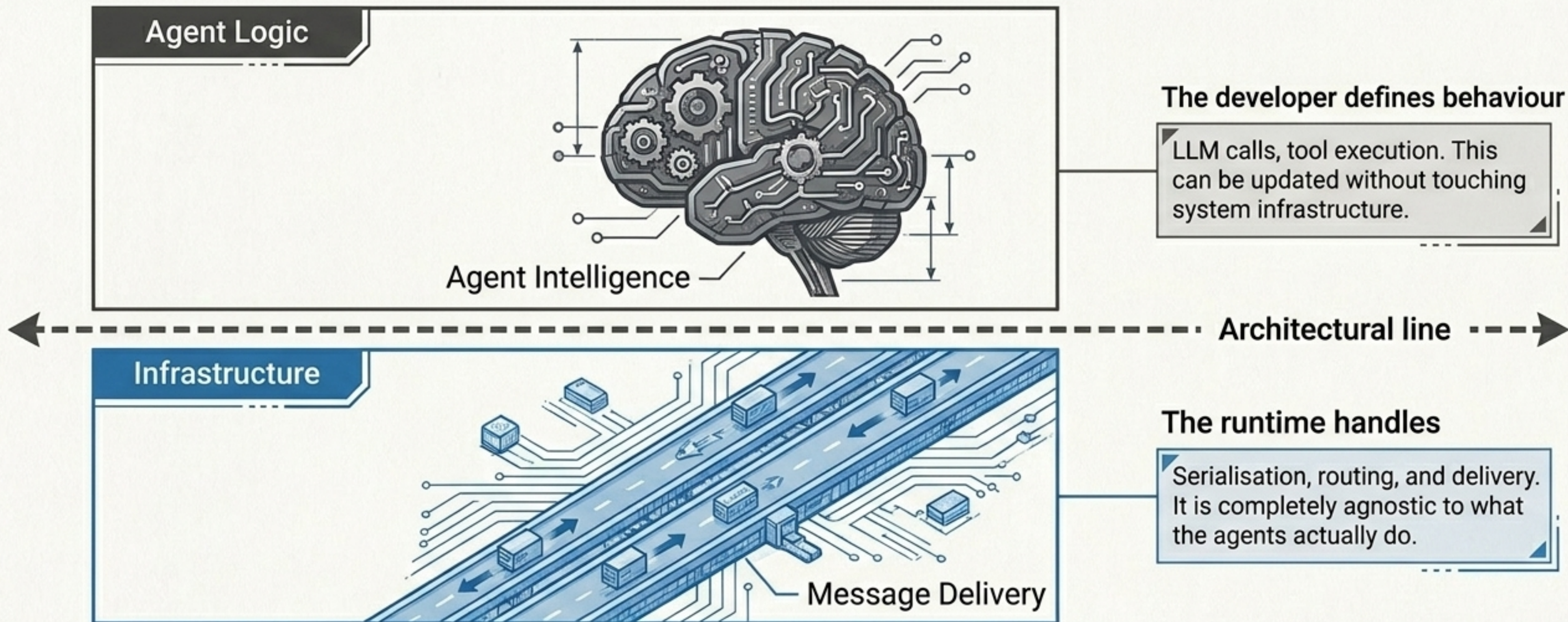
Decoupled Logic

Asynchronous and
Event-Driven

The Layered Ecosystem Blueprint



AutoGen Core: The Engine Room



Location Transparency

Agents interact identically whether running locally in-process or distributed across a remote network.

Wiring the Agents: Identity and Routing

Declarative Routing

Registers specific methods to intercept specific typed messages automatically.

Type Safety

Agents publish and subscribe to strictly typed data objects, not direct method calls.

```
1 @message_handler
2 async def on_research_request(self, message: ResearchRequest, ctx: MessageContext):
3     # Agent behaviour implemented here
4     pass
```

AgentId

AgentId

[Type Label: 'research_agent']

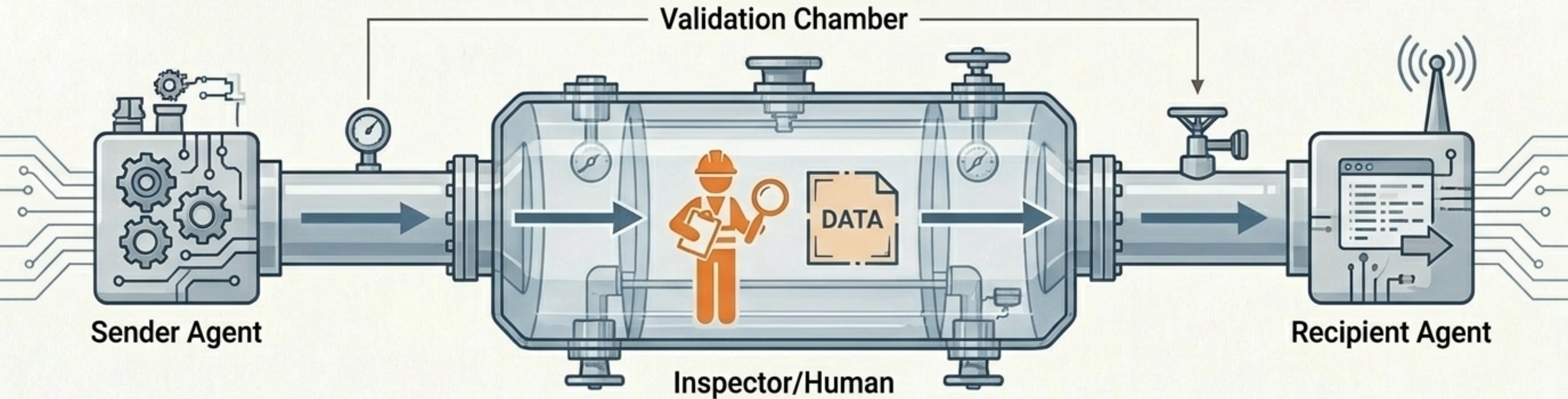
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Globally Unique AgentId

Middleware: The Transparent Pipeline



Transparent Interception

Sits securely between the message bus and agent handlers without the agents knowing.

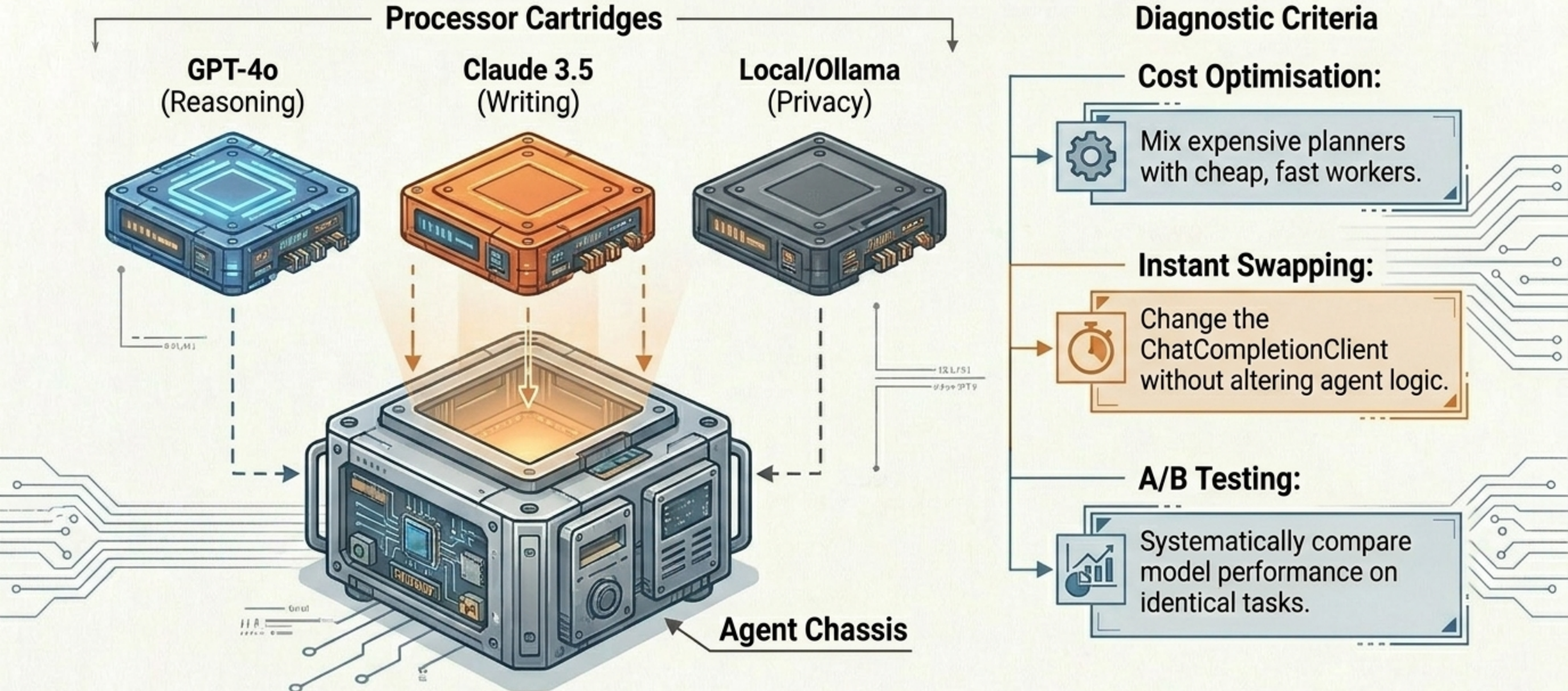
Human-in-the-loop

Intercepts messages for seamless human review, approval, or protective rate limiting.

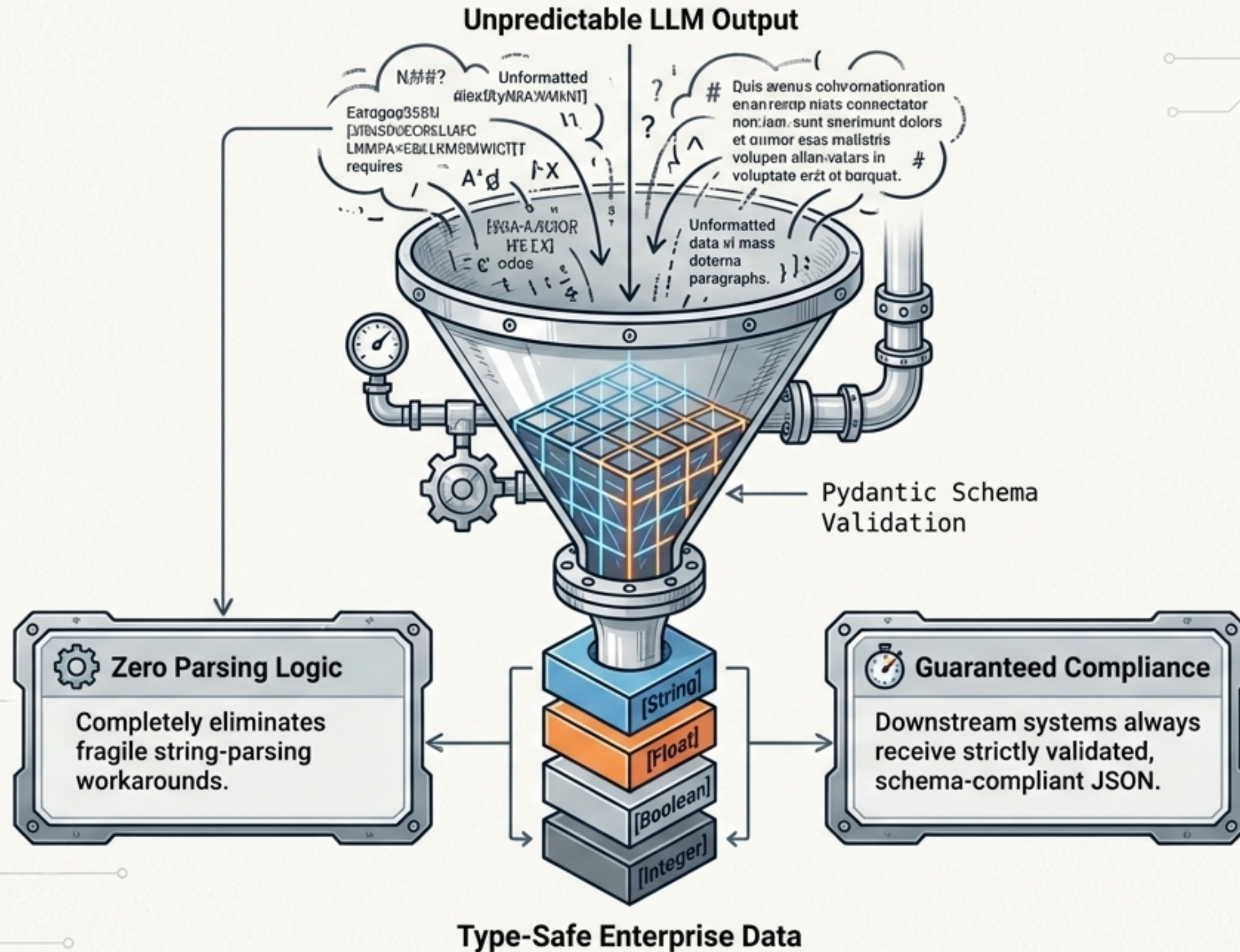
Global Observability

Logs and traces all inter-agent communications for comprehensive auditing.

Multi-Model Support: Brain Diversity

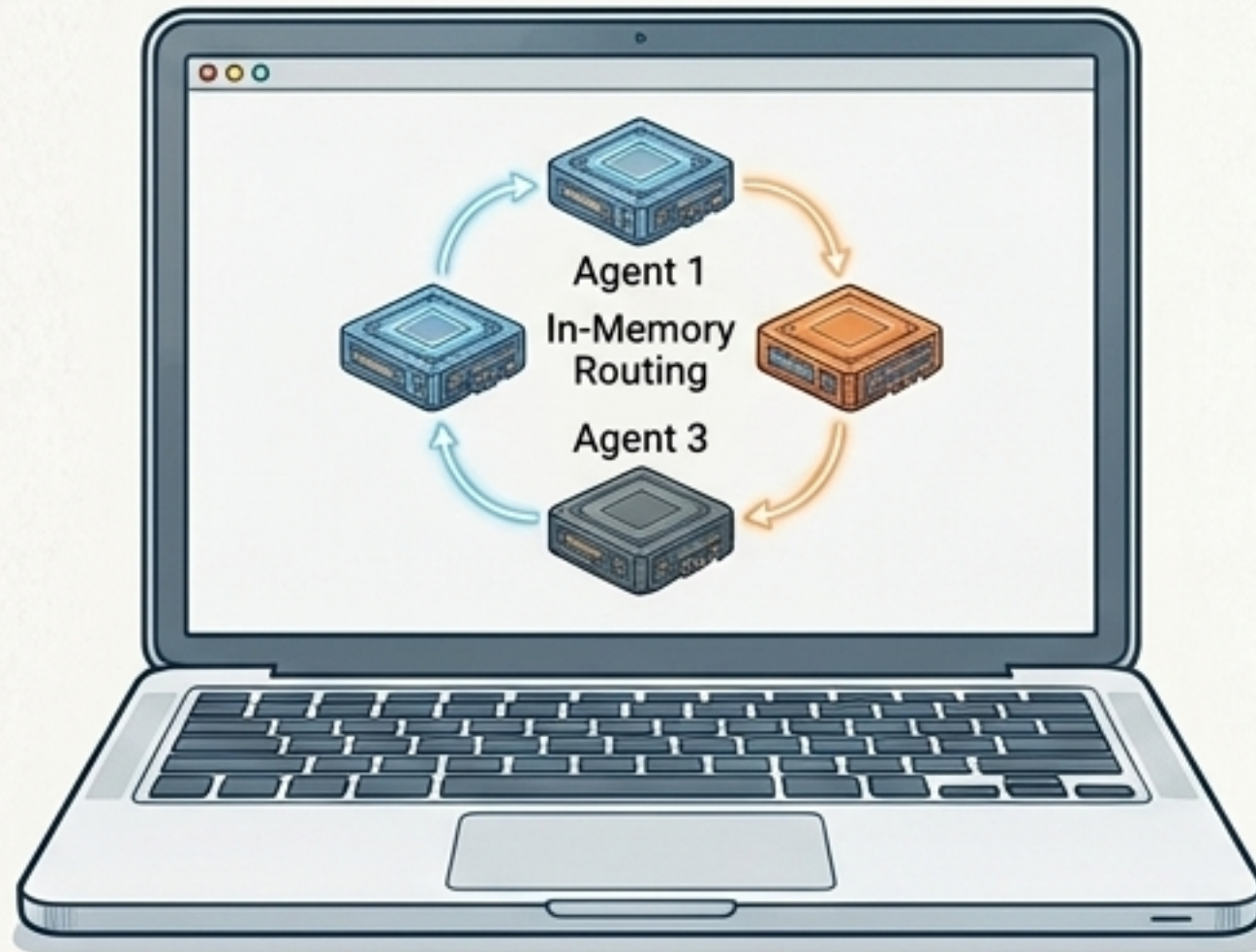


Structured Output: Taming the LLM



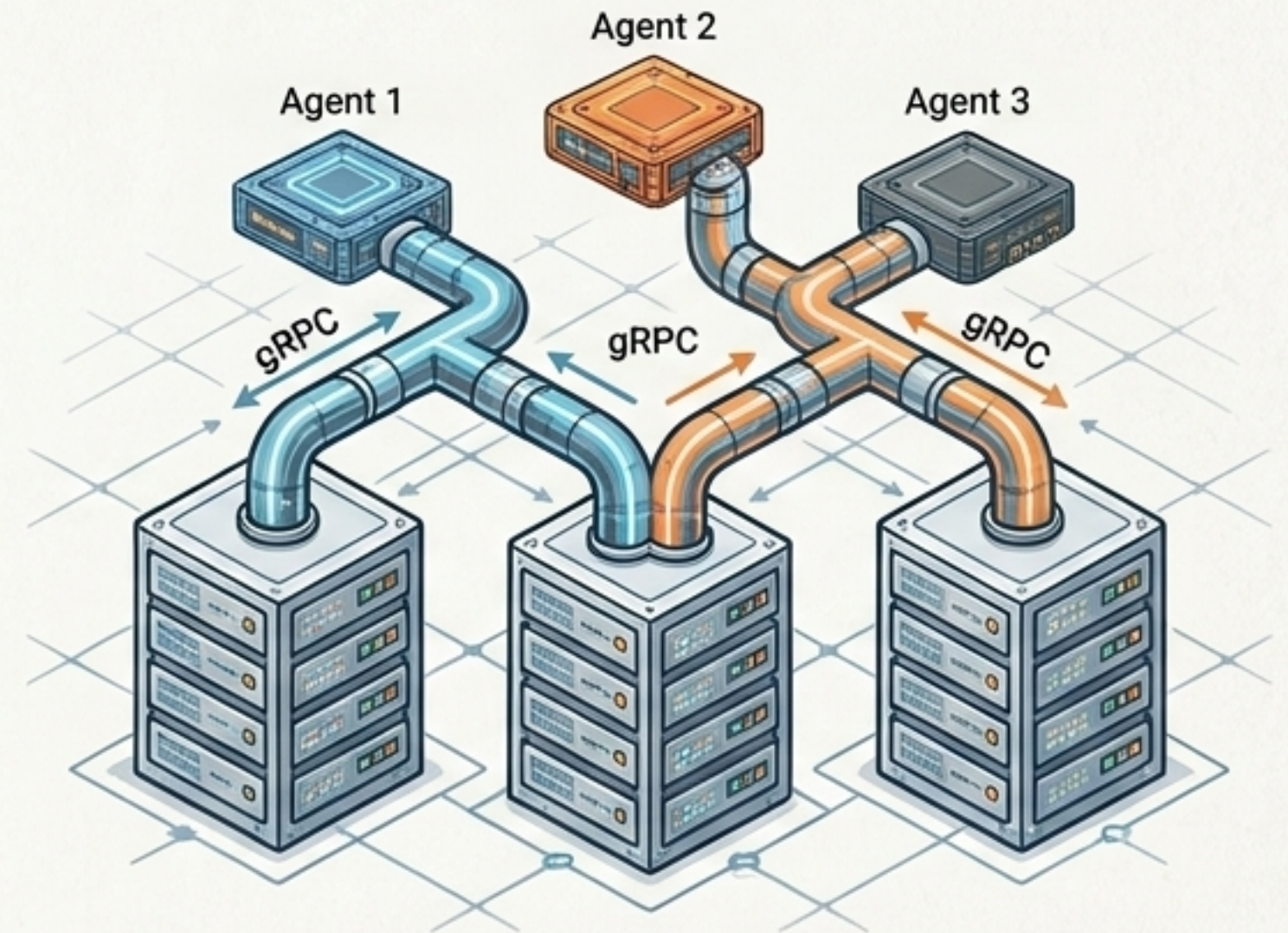
Distributed Messaging: Scaling Up

Single-Process Runtime



SingleThreadedAgentRuntime for rapid local testing and iteration.

Distributed Cloud Network



GrpcWorkerAgentRuntimeHost deploys agents across distributed host machines.



Note: Distributed Runtime capabilities are currently marked as experimental.

AgentChat: Building Conversational Teams

AssistantAgent

Model Client

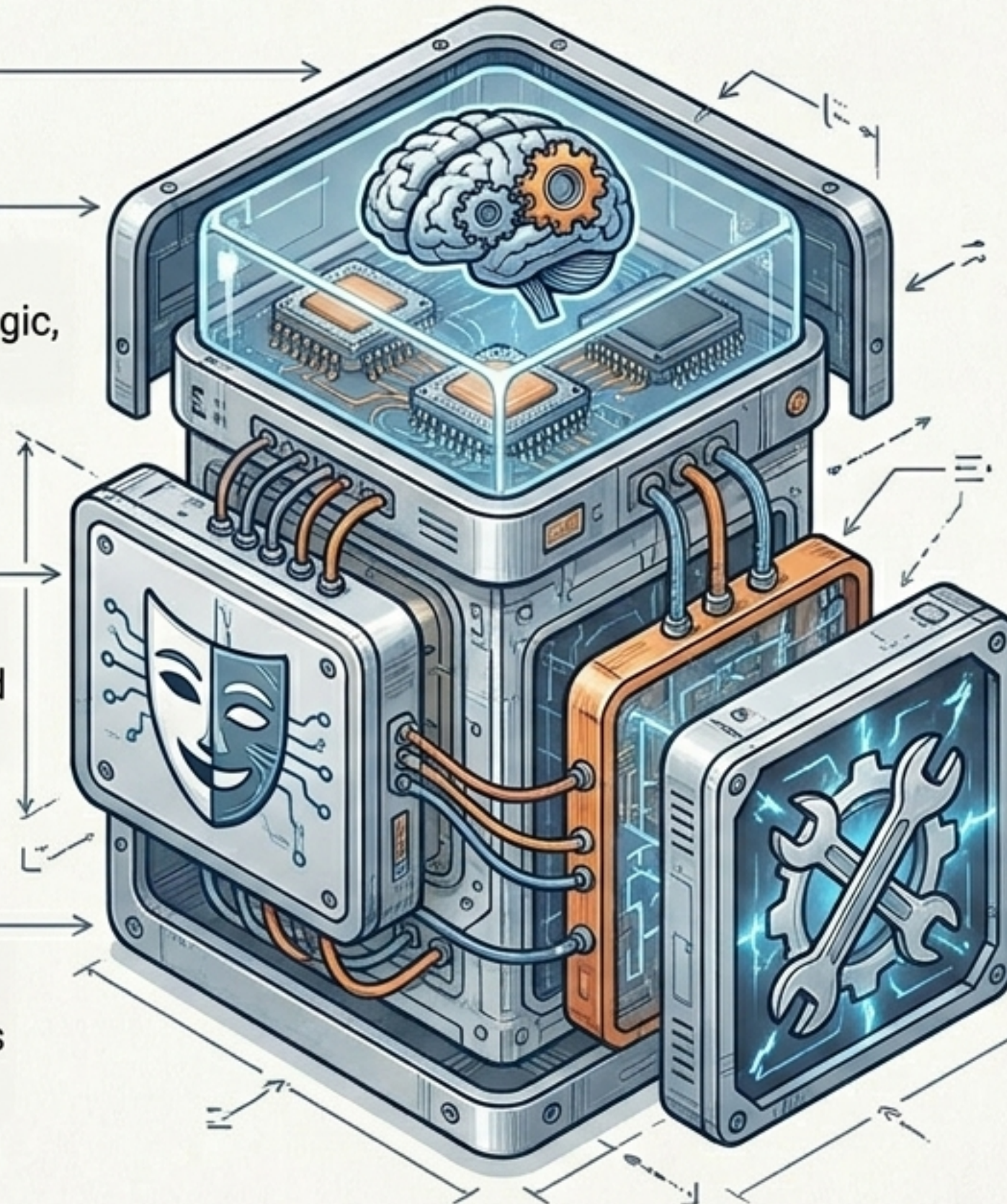
Handles all LLM API communications, retry logic, and token streaming.

System Message

Defines the agent's persona, constraints, and operational guidelines.

Tools & Functions

Equips the agent with custom Python functions or external REST APIs.

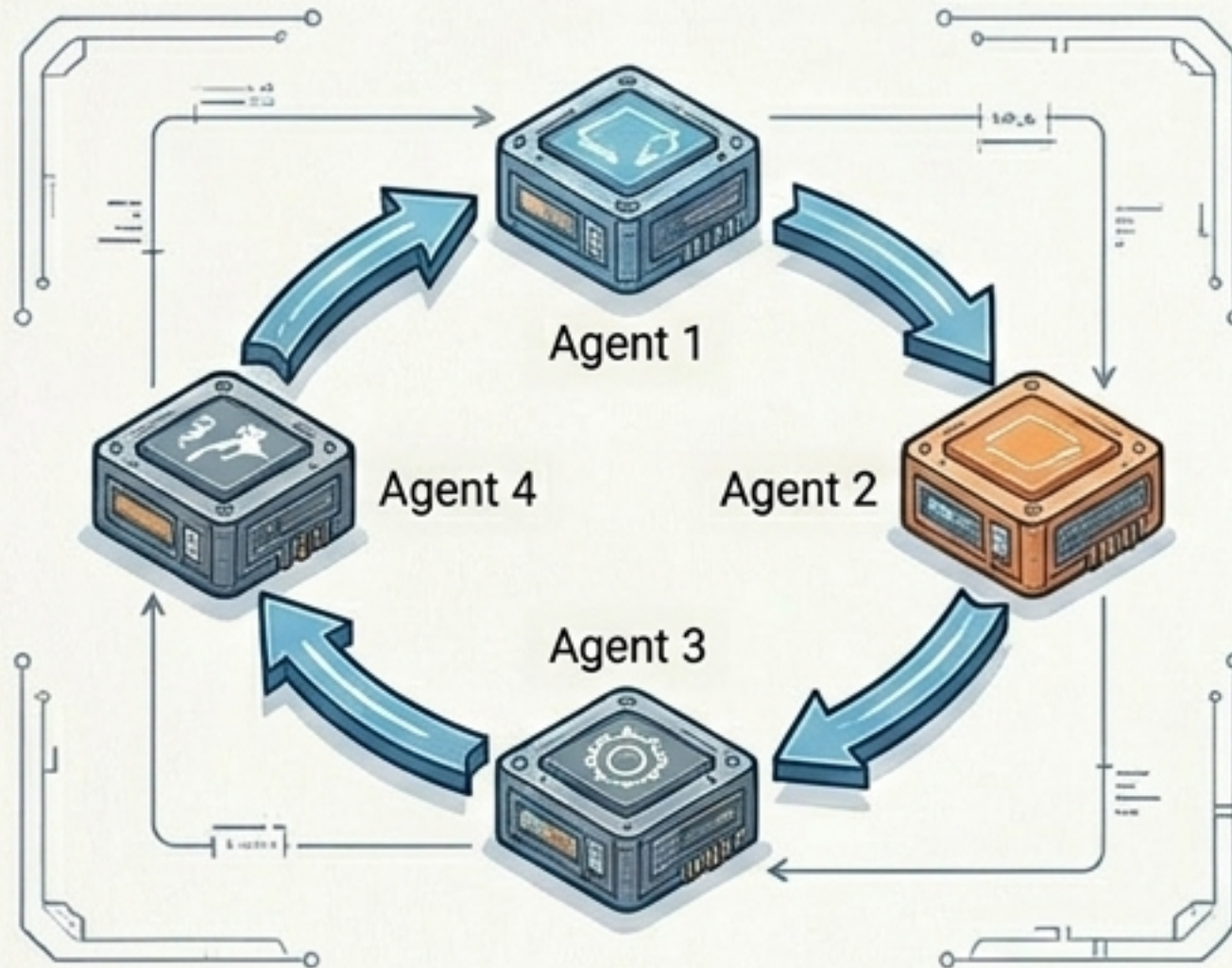


Communication Standard:

Relies entirely on strongly-typed objects like TextMessage and ToolCallMessage.

Orchestrating Team Dynamics

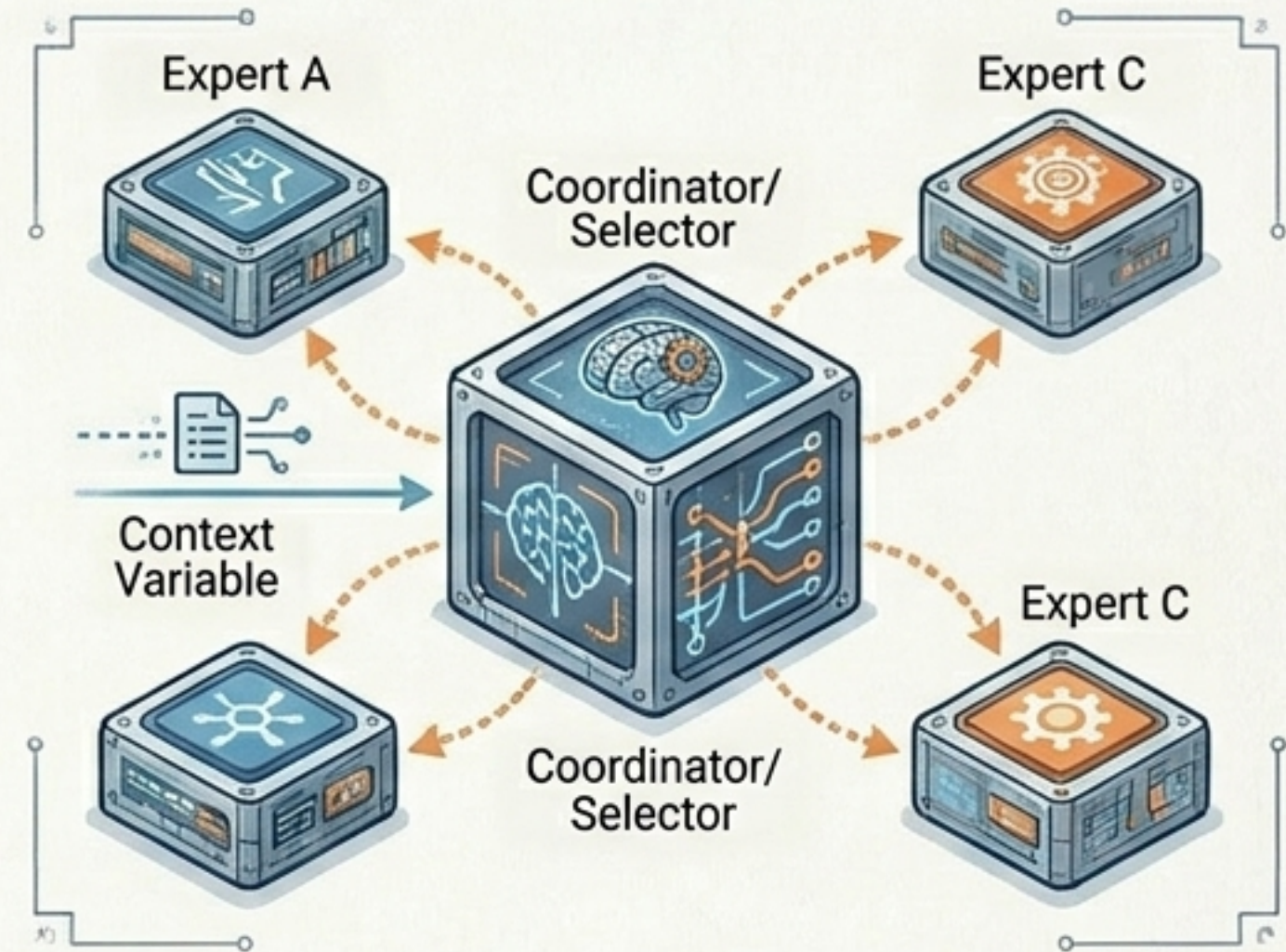
RoundRobinGroupChat



Predictable Turn-Taking

- Agents respond in a fixed, sequential order. Ideal for strict pipeline tasks where each expert must review the previous step.

SelectorGroupChat



Dynamic LLM Routing

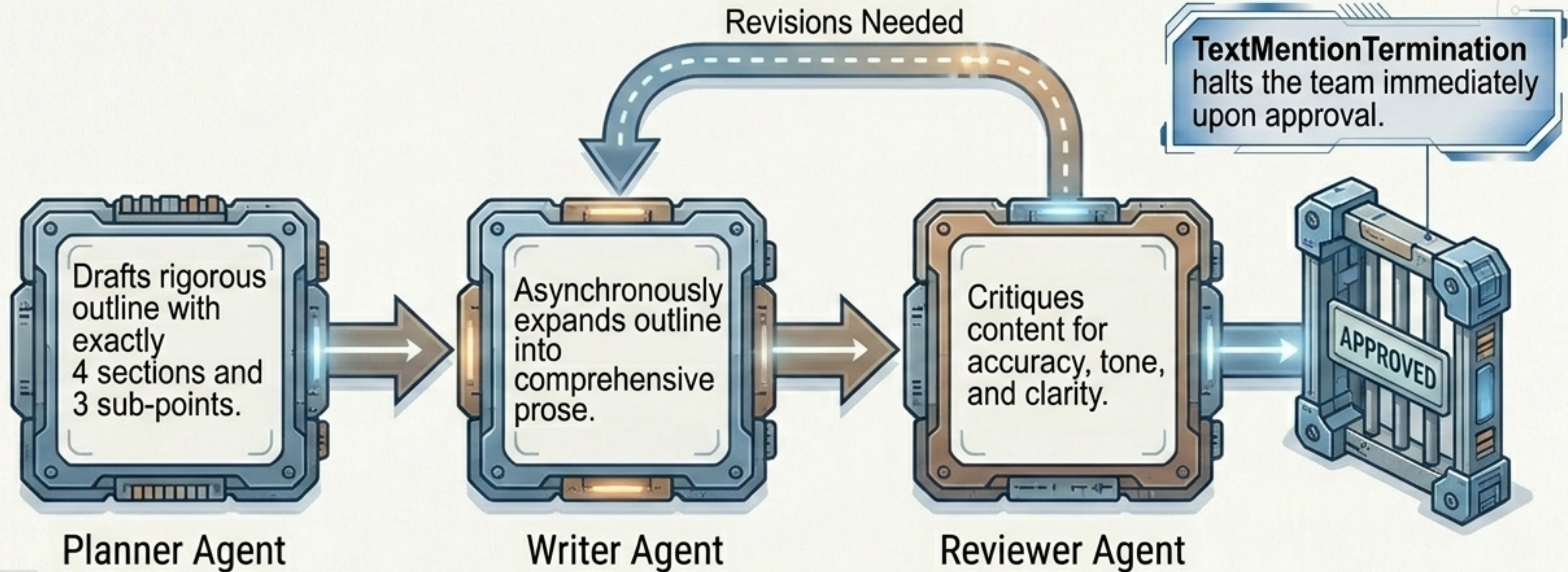
- A designated selector agent evaluates conversation history and dynamically chooses the most appropriate expert to speak next.

Architectural Decisions: AutoGen vs. Semantic Kernel

Dimension	AutoGen 0.4	Semantic Kernel
Primary Focus	Multi-agent orchestration and dynamic team communication.	Integrating AI capabilities directly into existing applications.
Core Abstraction	Autonomous agents passing typed messages.	Deterministic plugins invoked by a central kernel.
Best Use Case	Complex workflows requiring task decomposition and debate.	Enhancing existing .NET/Python codebases with AI functions.

Highly Complementary: AutoGen teams can seamlessly trigger Semantic Kernel plugins as registered tools.

In Action: Content Creation Workflow



The Blueprint Realised

Infinite Scale

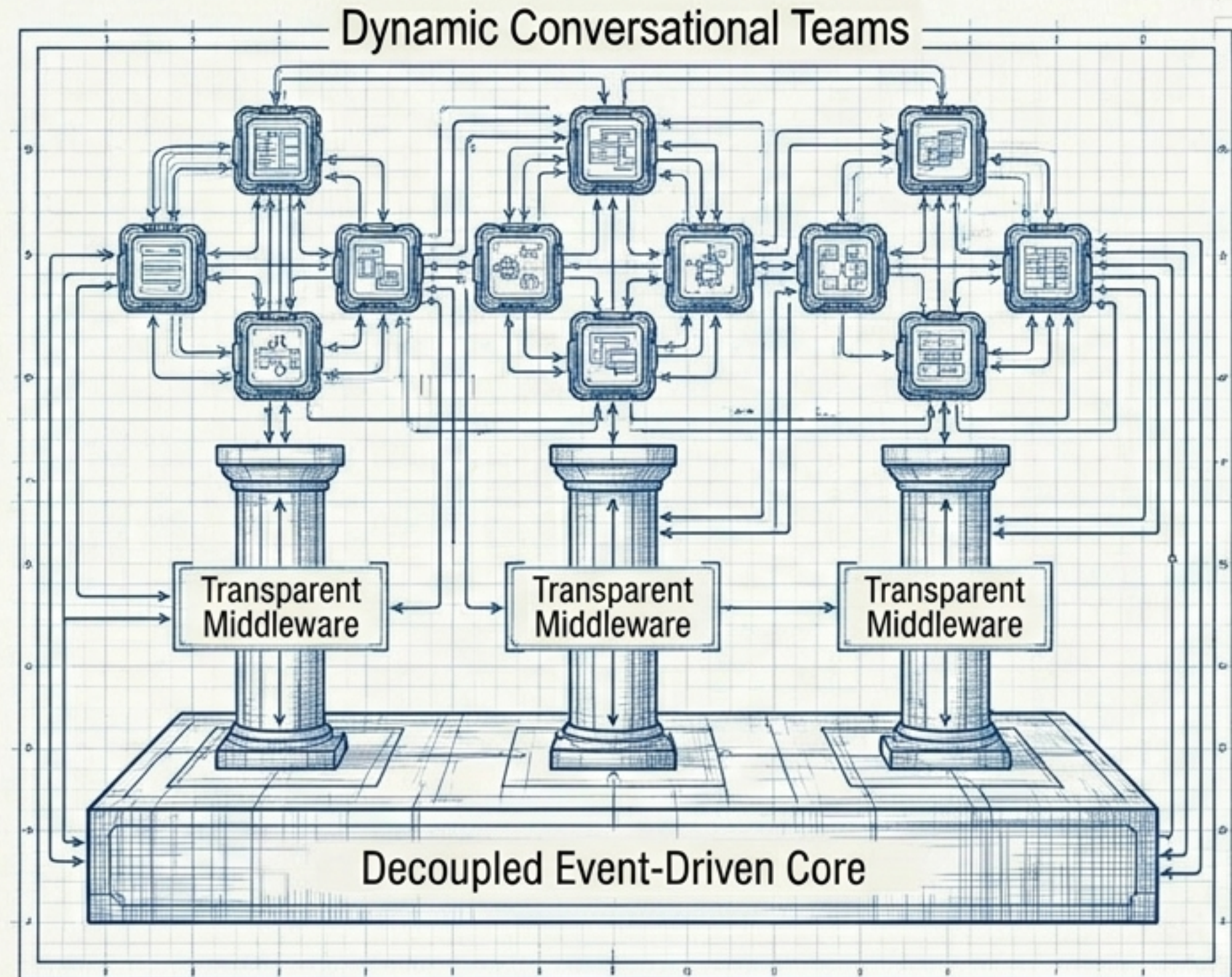
Strict decoupling guarantees that logic, routing, and models evolve independently.

Event-Driven

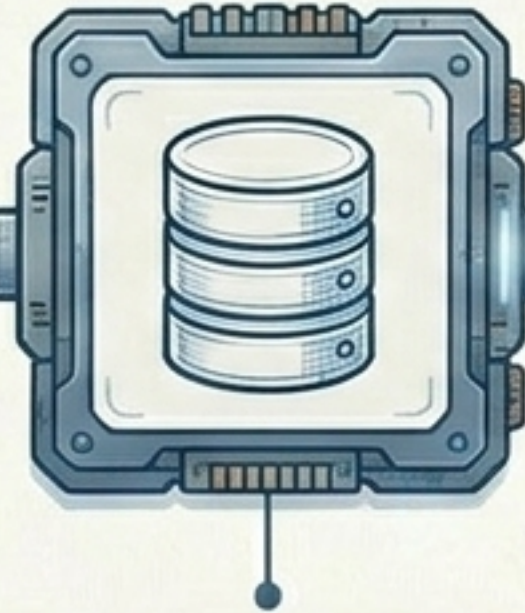
Replaces rigid conversational scripts with dynamic, asynchronous networks.

Enterprise Ready

Delivers a truly modular foundation for production-grade AI systems.



Beyond the Blueprint: What's Next



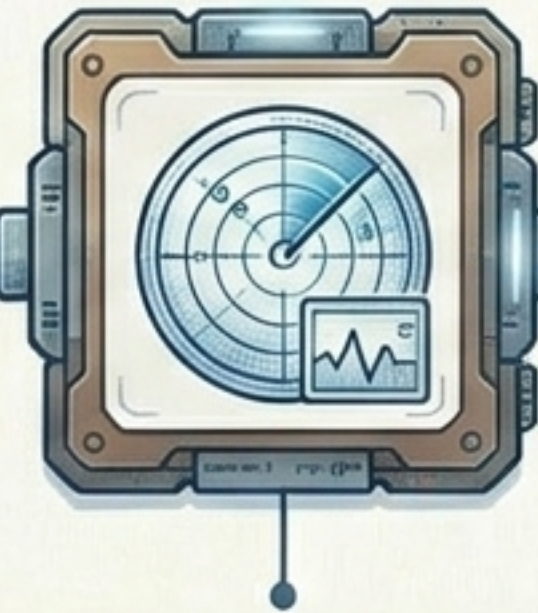
Memory Systems

Integrating external vector stores (ChromaDB, Qdrant) for long-term agent recall and semantic search.



Tool Integration

Equipping agents with custom REST APIs, Python code execution, and Semantic Kernel plugins.



Observability

Tracking complex message flows via OpenTelemetry and structured JSON logging.

Next Module: Advanced Distribution across Global Cloud Networks.